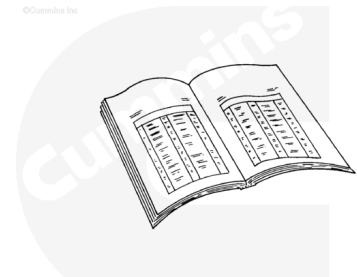


Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



ck800wa

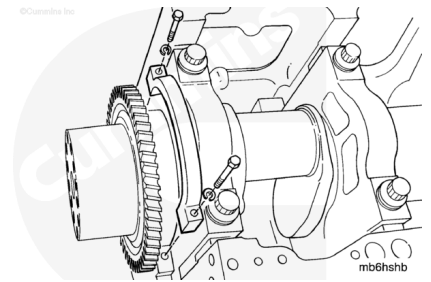
⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Disconnect the batteries. Refer to the OEM service manual.
- Drain the engine oil. Refer to Procedure 007-025 in Section. (</qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html>)
- Remove the oil filters. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html>)
- Remove the oil pan adapter assembly from the block flange. Refer to Procedure 007-027 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html>)
- Remove all oil transfer and suction tubes and brackets. Refer to Procedure 007-035 in Section 7. (</qs3/pubsys2/xml/en/procedures/28/28-007-035-tr.html>)
- Remove all oil transfer and suction tubes and brackets. Refer to Procedure 001-061 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-061.html>)

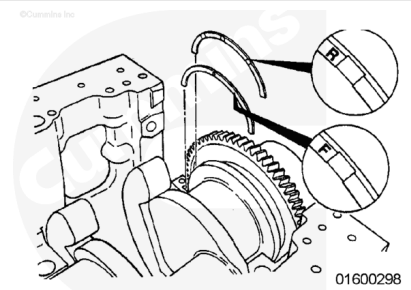
Remove

Remove the two capscrews from the lower thrust bearing retainer and remove the retainer.



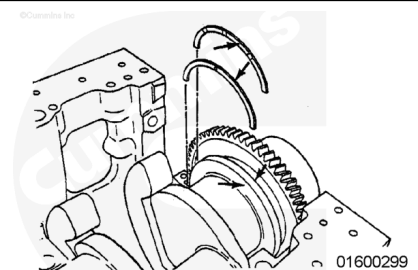
To remove the thrust bearing halves, keep the crankshaft at one end of its thrust and rotate it using the barring mechanism. If **only** one of the bearing halves rotates out, move the crankshaft to the other end of its thrust and rotate the crankshaft again.

Mark the location of each thrust bearing, front and rear, on the side opposite the one which faces the crankshaft flange.



Inspect the faces on the crankshaft thrust flange and the thrust surfaces of the bearing halves. Thrust bearings are available in standard and the following oversizes:

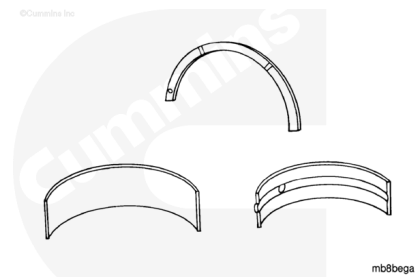
- 0.254 mm [0.010 in]
- 0.508 mm [0.020 in]
- 0.762 mm [0.030 in]
- 1.016 mm [0.040 in]



Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

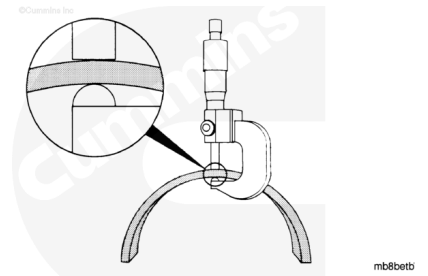
⚠ CAUTION ⚠

Do not use a scraper or a wire brush. This can cause damage to the bearings.

Use solvent to clean the bearings.

Dry with compressed air.

Use a micrometer to measure the thrust bearing thickness at the wear location. The shine indicates where the crankshaft has contacted the bearing.



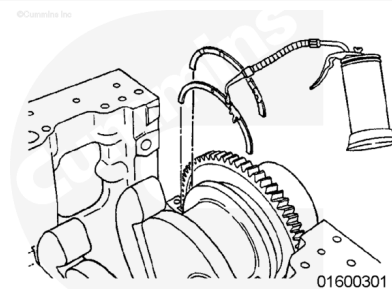
Thrust Bearing Thickness Standard or Oversize			
	mm		in
Standard	4.826	MIN	0.1900
	4.940	MAX	0.1945
Oversize 0.254 mm [0.010 in]	5.080	MIN	0.2000
	5.194	MAX	0.2045
Oversize 0.508 mm [0.020 in]	5.334	MIN	0.2100
	5.448	MAX	0.2145
Oversize 0.762 mm [0.030 in]	5.588	MIN	0.2200
	5.702	MAX	0.2245
Oversize 1.016 mm [0.040 in]	5.842	MIN	0.2300
	5.956	MAX	0.2345

Install

Note : The grooves in the thrust bearings must face the flange on the crankshaft.

Lubricate the two thrust bearings with clean engine oil and insert them into the thrust bearing support.

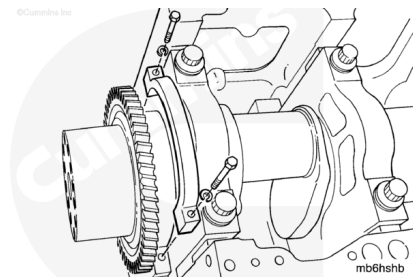
Note : Move the crankshaft to the front or rear as necessary to insert the second thrust bearing.



Note : The machined side of the lower thrust bearing retainer must be toward the main bearing cap.

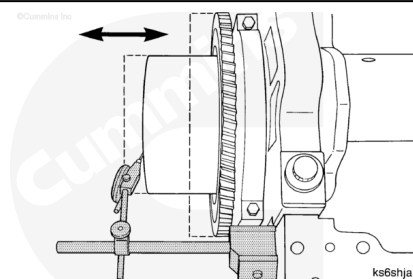
Install the thrust bearing retainer, lock washer, and capscrews.

Torque Value: 40 n•m [30 ft-lb]



Use a dial indicator to measure end clearance. Move the crankshaft to one end of its thrust and zero the dial. Move the crankshaft to the other end of its thrust and take the reading.

Crankshaft End Clearance		
mm		in
0.13	MIN	0.005
0.51	MAX	0.020



If the clearance is **not** within specifications, check for foreign material on the thrust bearings or crankshaft flange. Oversize thrust bearings are available to adjust the end clearance.

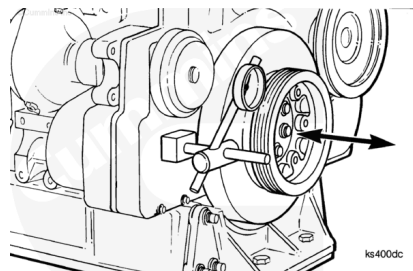
Note : It is recommended to recheck the crankshaft end clearance once the engine is installed into an application.

Release the tension on the cooling fan drive belt, if installed. Refer to Procedure 008-002 in Section 8.

(/qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html)

Remove an accessible hand hole cover. Refer to Procedure 001-035 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-035-tr.html)



Use the barring device to rotate the engine so that the connecting rod journal is opposite the hand hole cover opening. If the connecting rods are already in this position when the hand hole cover is removed, use the barring device to rotate the crankshaft slightly in the opposite direction to normal rotation. This releases the loading on the gear train.

Attach a dial indicator, resting against the vibration damper.

Use a pry bar or similar to move the crankshaft backwards by prying against the crankshaft web.

Note : Do not pry against the connecting rods or crankshaft counterweights.

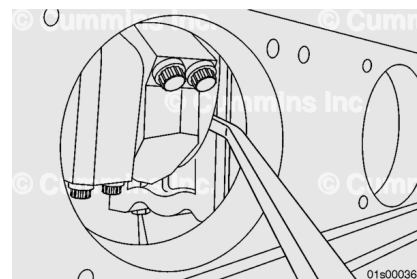
Set the dial indicator to zero.

Move the crankshaft in the opposite direction by prying on the crankshaft web.

Check the reading on the dial indicator to confirm the crankshaft has end clearance, which is within the limits shown above.

If previously loosened, tension the cooling fan drive belt. Refer to Procedure 008-002 in Section 8.

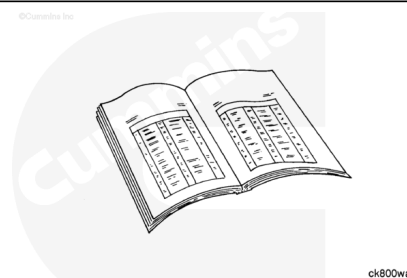
(/qs3/pubsys2/xml/en/procedures/28/28-008-002-tr.html)



Finishing Steps

⚠ WARNING ⚠

This component weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install oil transfer and suction tubes and brackets. Refer to Procedure 007-035 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-007-035-tr.html)
- Install oil transfer and suction tubes and brackets. Refer to Procedure 001-061 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-061.html)
- Install the oil pan adapter assembly to the block flange using a new gasket. Refer to Procedure 007-027 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-027-tr.html)
- Install new oil filters. Refer to Procedure 007-013 in Section 7. (/qs3/pubsys2/xml/en/procedures/28/28-007-013-tr.html)
- Fill the engine with clean 15W-40 engine oil. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/28/28-007-025-tr.html)
- Connect the batteries. Refer to the OEM service manual.
- Operate the engine and check for leaks.